Algebra Homework 3

Problem 1. Let F be a field and $K = F(X)$.

(i) Determine the algebraic closure of F in K.

(ii) Given $b \in K$ transcendental over F, find a valuation ring of K that does not contain b.

Proof. (i) The algebraic closure of F in K is F itself. To see this, take $b = p(X)/q(X) \in K \setminus F$, where $p, q \in F[X]$ and $q \neq 0$. The polynomial $p(T) - bq(T) \in F(b)[T]$ is nonzero otherwise a nonzero coefficient of q would give $b \in F$. Since it vanishes at X, the element X is algebraic over $F(b)$. Assume that b is algebraic over F. The transitivity would give X is algebraic over F, which is a contradiction.

(ii) Write $b = p(X)/q(X)$ with p, q coprime. If q is nonconstant, choose an irreducible factor r of q and define

$$B := F[X]_{(r)} = \left\{ \frac{g}{h} : g, h \in F[X],\ r \nmid h \right\}.$$

This is a subring of K. For any $z = g/h \in K^\times$ with g, h coprime, either $r \nmid h$, so $z \in B$, or $r \mid h$ and hence $r \nmid g$, so $z^{-1} \in B$. Thus B is a valuation ring by Definition 75. Moreover, $b \notin B$, since $r \mid q$ and $r \nmid p$.

If q is constant, then b is a nonconstant polynomial. In this case define

$$B_\infty := \{0\} \cup \left\{ \frac{g}{h} : g, h \in F[X] \setminus \{0\},\ \deg g \leq \deg h \right\}.$$

Then B_∞ is again a subring of K. Every $z = g/h \in K^\times$ satisfies either $\deg g \leq \deg h$ or $\deg h \leq \deg g$, so $z \in B_\infty$ or $z^{-1} \in B_\infty$. Hence B_∞ is a valuation ring. Since $\deg b > 0$, it does not contain b. □

Problem 2. Prove Theorem 72 without assuming that $\ker \varphi$ is maximal.

Proof. Let B/A be integral, let $\mathfrak{q} \in \operatorname{Spec}(B)$, and put $\mathfrak{p} = \mathfrak{q} \cap A$. Suppose $\varphi : A \to L$ is a homomorphism to a field with $\ker \varphi = \mathfrak{p}$. Then there exists a homomorphism $\Phi : B \to \overline{L}$ extending φ and satisfying $\ker \Phi = \mathfrak{q}$.

Set $S = A - \mathfrak{p}$. Localization gives an integral extension $S^{-1}B/A_\mathfrak{p}$, and the prime ideal $\mathfrak{q}' = S^{-1}\mathfrak{q}$ contracts to the maximal ideal $\mathfrak{p}A_\mathfrak{p}$. For completeness, integrality is preserved because a monic equation

$$b^n + a_1 b^{n-1} + \cdots + a_n = 0$$

over A yields a monic equation for b/s over $A_\mathfrak{p}$ with coefficients a_i/s^i.

Since $\varphi(s) \neq 0$ for all $s \in S$, the map φ extends to

$$\varphi_\mathfrak{p} : A_\mathfrak{p} \longrightarrow L, \qquad \frac{a}{s} \longmapsto \frac{\varphi(a)}{\varphi(s)},$$

whose kernel is $\mathfrak{p}A_\mathfrak{p}$. Theorem 72 now applies and gives $\Psi : S^{-1}B \to \overline{L}$ with $\Psi|_{A_\mathfrak{p}} = \varphi_\mathfrak{p}$ and $\ker \Psi = \mathfrak{q}'$. Define $\Phi(b) = \Psi(b/1)$. Then $\Phi|_A = \varphi$, and

$$\ker \Phi = \{b \in B : b/1 \in S^{-1}\mathfrak{q}\} = \mathfrak{q}.$$

The last equality holds because $\mathfrak{q}$ is prime and $S \cap \mathfrak{q} = \varnothing$. □

Problem 3. Let $|\cdot| : K \to \mathbb{R}_{\geq 0}$ be a norm and $\mathfrak{o}_{|\cdot|} = \{x \in K : |x| \leq 1\}$. Prove that $\mathfrak{o}_{|\cdot|}$ is a valuation ring if and only if $|\cdot|$ is non-Archimedean.

Proof. Suppose first that $\mathfrak{o}_{|\cdot|}$ is a valuation ring. If x and y are not both zero, interchange them if necessary so that $|x| \geq |y|$ and $x \neq 0$. Then $y/x \in \mathfrak{o}_{|\cdot|}$. Since $\mathfrak{o}_{|\cdot|}$ is a subring, $1 + y/x \in \mathfrak{o}_{|\cdot|}$, and hence

$$|x + y| = |x| \left|1 + \frac{y}{x}\right| \leq |x|.$$

It is immediate that $x = y = 0$.

Conversely, suppose that $|\cdot|$ is non-Archimedean. Multiplicativity gives $|1| = |-1| = 1$, so $\mathfrak{o}_{|\cdot|}$ contains 0 and 1 and is closed under additive inverses and multiplication. For any $x, y \in \mathfrak{o}_{|\cdot|}$, the triangular inequality gives

$$|x + y| \leq \max\{|x|, |y|\} \leq 1.$$

Thus $\mathfrak{o}_{|\cdot|}$ is a subring of K. If $x \in K^\times$ does not belong to $\mathfrak{o}_{|\cdot|}$, then $|x| > 1$, so $|x^{-1}| = |x|^{-1} < 1$. Hence $x^{-1} \in \mathfrak{o}_{|\cdot|}$, proving that $\mathfrak{o}_{|\cdot|}$ is a valuation ring. □

Problem 4. Can every valuation ring be described by a non-Archimedean norm with values in $\mathbb{R}_{\geq 0}$?

No. The following examples were collected from students' homework solutions. They are listed without proofs. Please check for yourselves that, in each case, B is a valuation ring of K that cannot be described by a non-Archimedean norm with values in $\mathbb{R}_{\geq 0}$.

Here k denotes any field, and

$$\mathbb{Z}_{(2)} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z},\ 2 \nmid b \right\}.$$

(i) Let
$$K = k(X, Y), \qquad V = k(Y)[X]_{(X)}, \qquad B = k[Y]_{(Y)} + XV.$$

(ii) Let
$$K = k((X))((Y)), \qquad V = k((X))[[Y]], \qquad B = k[[X]] + YV.$$

(iii) Let
$$K = \mathbb{Q}(X), \qquad V = \mathbb{Q}[X]_{(X)}, \qquad B = \mathbb{Z}_{(2)} + XV.$$

(iv) Let
$$K = \mathbb{Q}((T)), \qquad V = \mathbb{Q}[[T]], \qquad B = \mathbb{Z}_{(2)} + TV.$$

Problem 5. Is there a field K with two proper valuation rings $B_1 \subsetneq B_2$?

Yes. Each example collected in Problem 4 also gives an example for this question. Take

$$B_1 = B, \qquad B_2 = V,$$

and it follows that

$$B_1 \subsetneq B_2 \subsetneq K.$$